On the phase stability of high entropic oxide catalysts for the hydrogenation of CO_2 to methanol

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Expanding the concept of high entropy from alloys to oxides opened a new avenue for tailoring and stabilizing heterogeneous catalysts and or catalyst support materials [1]. High entropic oxides (HEO) are particularly attractive due to their inherent propensity towards oxygen vacancies which are crucial in the activation of CO_2 for the hydrogenation to methanol [2,3]. Moreover, it was shown that utilizing Cu in HEOs can lead to a stabilization of Cu(II) and Cu(I) oxidation states under reductive conditions [4-6]. The utilization of multiple oxidation states of Cu was found beneficial in the hydrogenation of CO_2 to methanol by unlocking a mechanistic dual-channel pathway in the hydrogenation reaction [7].

HEOs adapt crystal structures in higher symmetry in comparison to the expected crystal structure for the individual principal components it consists of. For Cu based systems the monoclinic tenorite phase and its propensity towards reversible dissolution into a crystal structure of higher symmetry is often regarded as a benchmark for the entropic stabilization effect [8]. Moreover, Cu^{2+} featuring octahedral geometry in HEOs can lead to Jahn Teller distortions, which stabilization effect can be thermally overcome in HEOs at temperatures relevant for CO_2 hydrogenation to methanol [5,9].

In this work, the rich structural diversity when employing copper in HEOs is investigated for $AlCeCuZnZrO_x$ and $AlCeCuTiZnZrO_x$ and B-doped variants of both candidate catalysts.

At specific synthesis conditions, the solubility limit of CuO (tenorite) in fluorite type $AlCeCuZnZrO_x$ HEO can be overcome by annealing at higher temperatures. In addition, first-order Jahn-Teller distortions are thoroughly investigated in the temperature regime relevant for methanol synthesis. Moreover, the reversible exsolution – dissolution of Cu^0 in the HEO is explored. These phenomena and their implications on the performance of HEO catalysts in the hydrogenation of CO_2 to methanol are studied in detail.

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